

ENTITY-RELATIONSHIP MODELING

Strong vs. Weak Entities

A Technical Comparison with Real-World Examples

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What is an Entity?

Building blocks of the ER Model

02

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Definition, key features & notation

03

Weak Entities

Definition, dependency & notation

04

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Applying the concepts to real systems

06

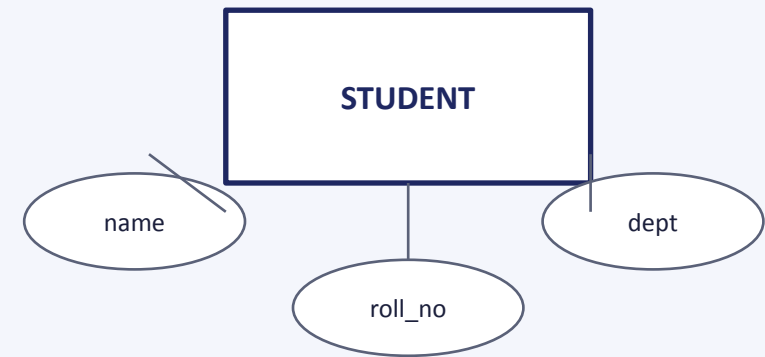
Summary

Key takeaways

What Is an Entity?

In ER (Entity-Relationship) modeling, an entity is a real-world object or concept — a person, place, event, or thing — that can be distinctly identified and stored as data in a database.

- Represented as a rectangle in an ER diagram
- Described by a set of attributes (properties)
- Every entity set has a related concept of identification
- Entities are broadly classified as Strong or Weak



Basic ER notation: entity + attributes

Strong Entity

A strong entity is an entity that can be uniquely identified by its own attributes, without depending on any other entity. It has its own primary key.

- + Exists independently of other entities
- + Has a primary key attribute
- + Represented as a single rectangle
- + Participates in relationships without depending on them for identity



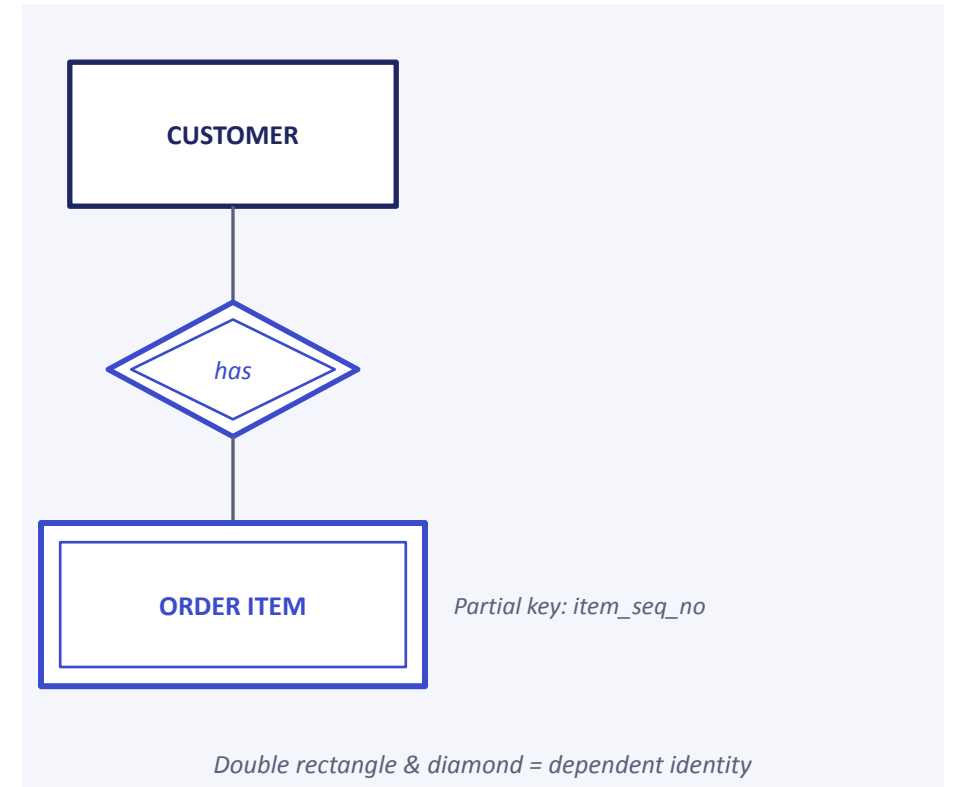
PK: customer_id

Single rectangle = independent identity

Weak Entity

A weak entity cannot be uniquely identified by its own attributes alone. It depends on a related "owner" (strong) entity through an identifying relationship.

- + Cannot exist without its owner entity
- + Has only a partial key (discriminator), not a full primary key
- + Represented as a double-outlined rectangle
- + Linked via a double-outlined diamond (identifying relationship)

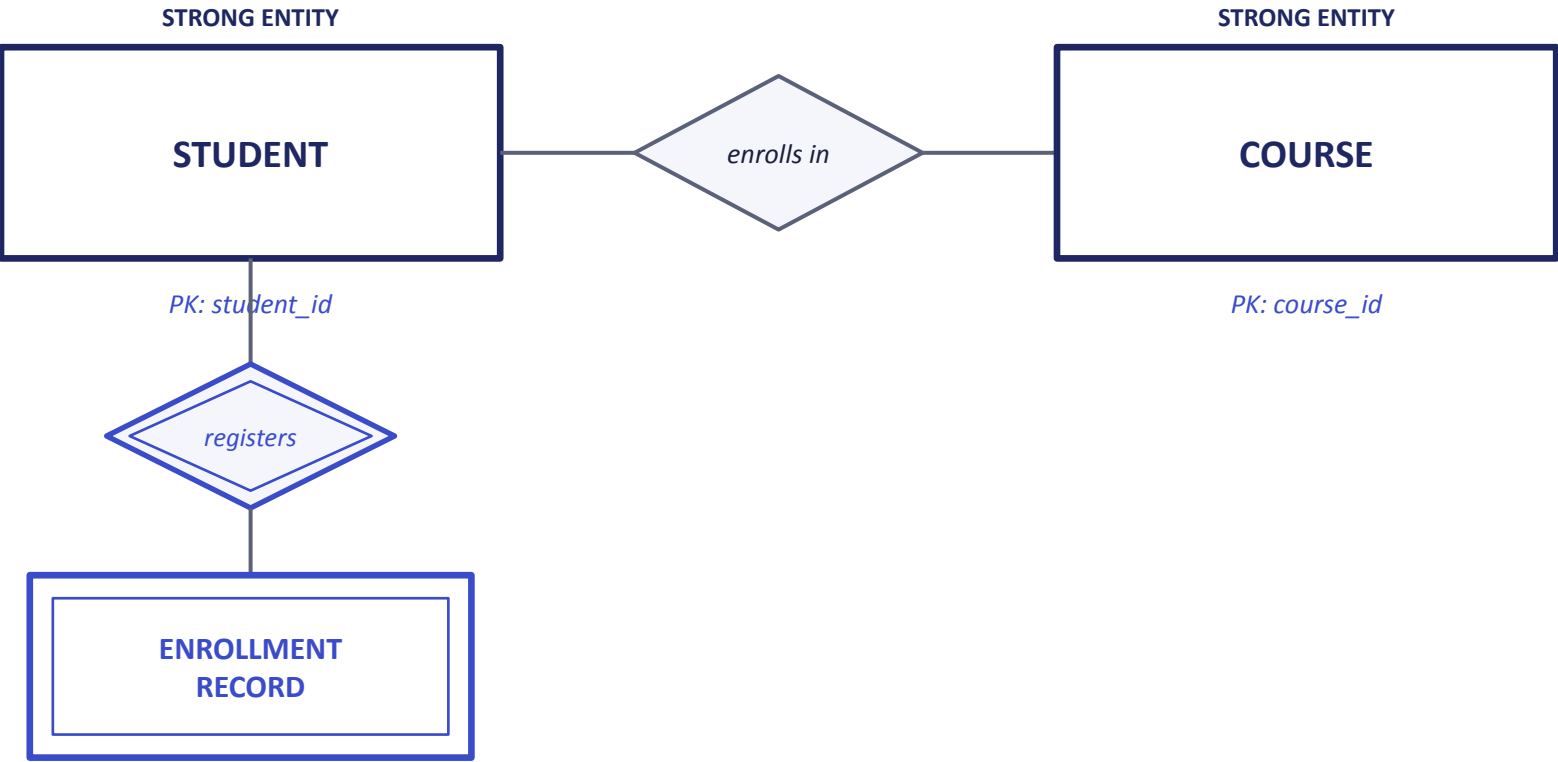


Strong vs. Weak Entities

Aspect	Strong Entity	Weak Entity
Identification	Uniquely identified by own attributes	Identified only via owner entity
Primary Key	Has its own primary key	Has only a partial key (discriminator)
Existence	Exists independently	Cannot exist without owner entity
ER Notation	Single rectangle	Double-outlined rectangle
Relationship	Regular relationship (diamond)	Identifying relationship (double diamond)
Example	Customer, Student, Employee	Order Item, Dependent, Room (in a Hotel)

Real-World Example: University System

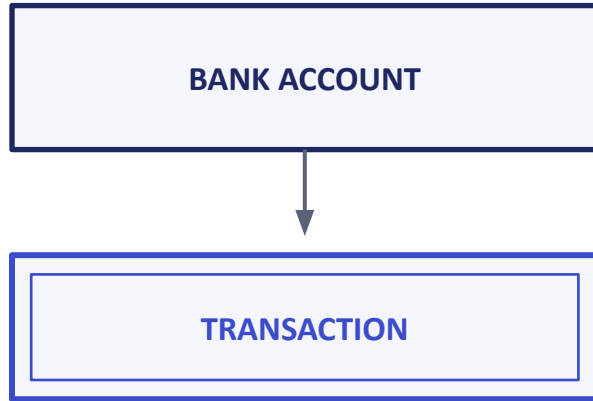
A STUDENT can enroll in multiple courses; each enrollment record cannot exist without a student.



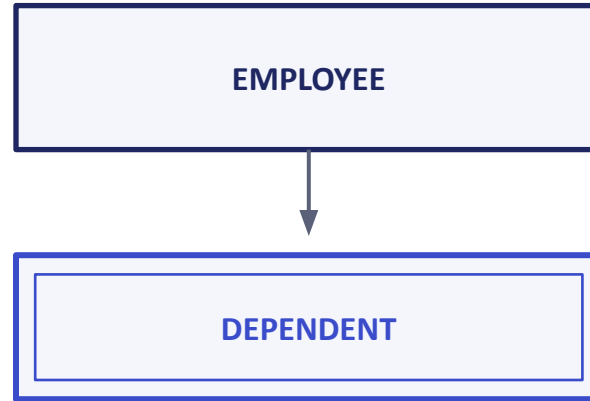
WEAK ENTITY
Enrollment Record

An enrollment record only makes sense in the context of a specific student — it uses student_id + semester as a partial key, and is deleted automatically if the student record is deleted.

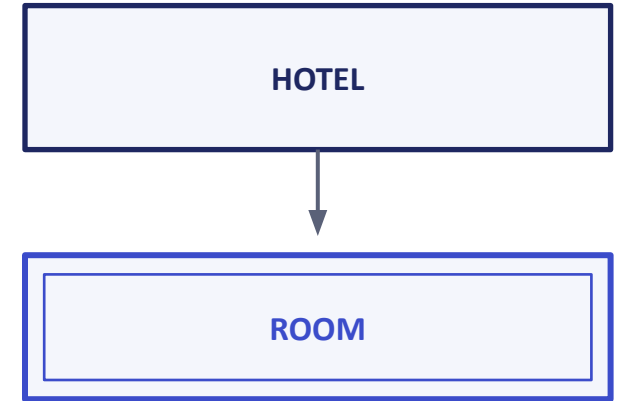
More Real-World Examples



A transaction record cannot exist without a linked account; it's identified by `account_no` + `txn_seq`.



A dependent (spouse/child) is meaningful only in relation to the employee; identified by `emp_id` + `dependent_name`.



Room numbers repeat across hotels; a room is identified only via `hotel_id` + `room_no`.

Key Takeaways

1

Independence

Strong entities stand alone with their own primary key; weak entities depend on an owner entity.

2

Notation

Strong = single rectangle & diamond. Weak = double rectangle & double diamond.

3

Design Impact

Correctly identifying strong vs. weak entities keeps the database normalized and avoids orphan records.

Thank You

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